

Each problem describes a conic section with a focus at the origin. Find the equation of the directrix associated with the focus at the origin, classify the conic section, and write the polar equation in standard form.

11) Eccentricity: 0.5

Vertices: $\left(0, \frac{4}{3}\right), (0, -4)$

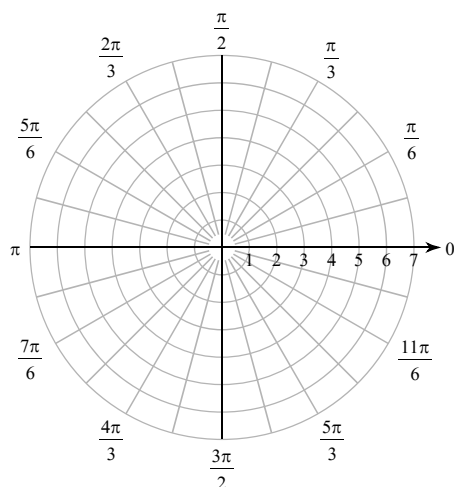
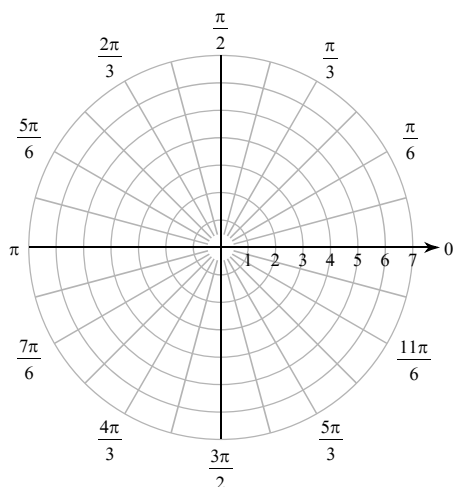
12) Eccentricity: 1

Vertex: $(-1, 0)$

Each polar equation describes a conic section with a focus at the origin. Find the eccentricity, find the equation of the directrix associated with the focus at the origin, classify the conic section, and graph the polar equation.

13) $r = \frac{19.8}{3 + 6.6\cos \theta}$

14) $r = \frac{9.6}{4 + 1.6\cos \theta}$



Critical thinking question:

15) Is it possible to use $r = \frac{ed}{1 + e\cos \theta}$ to represent a circle? Explain your answer.